

VBF-T9R4-DVPR-001 Design Verification Plan & Results

t9_r4 - grid-storage cell cathode on the Chen2020 NMC811/graphite profile

Result: HONEST NEGATIVE (no candidate satisfies the full contract; nothing reported as passing)

#	Requirement	Method	Result	Verdict
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1	avg voltage >= 4.6 V	run-comp x3 batches (42 layered)	max 4.6413 V (AIB55)	pass for AIB55 only
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2	charging potential <= 4.8 V	gate_charging.py incremental profile	7.410 V	FAIL (decisive)
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3	catalogue/hull exclusion	comp_envelope_check.py --formula/--point	PASS (outside envelope)	pass
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4	supported family	run-comp strict dispatch	layered LiMO2	pass
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5	ED >= 327.18 Wh/kg	run-pyamm 1C + calc-energy	not reached (no survivor)	n/a
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6	no plating 4C/45C	run-pyamm 4C_charge_45C --plating	not reached	n/a
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7	no TR trigger	run-tr	not reached	n/a
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8	SEI <= 500 nm @100cyc	run-pyamm aging_1C_100cyc	not reached	n/a
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9	true voltage >= 4.6 V	envelope re-compute + literature	4.6354 V independent re-compute	informative
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Verification closure: the mechanical audit chain is complete (log.jsonl: entry 0 criteria,

plan, 4 propose rounds each with same-round evaluate entries, 4 funnels, final entry).

close-loop: run-comp replay (envelope --formula re-computation, 4.6354 V) + public-catalogue

search (no results for LiAlO.5B0.5O2 / LiAlO2 cathodes - 'not found in the searched scope').

run-qe not performed: no candidate survived to Stage 5 (Al/B pseudopotentials were added to

the QE map in preparation; recorded honestly).